亞洲大學
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Midterm Project Report

Advanced Computer Programming

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Chapter 1 Introduction

1.1 Github

1) **Personal Github Account**: https://github.com/Linx25059
2) **Group Project Repository**: https://github.com/Linx25059/group-5

1.2 Overview

This Python script is a web scraper built with the **Scrapy** framework. It automatically visits a GitHub user's profile and extracts data from all their public repositories.

Key Features:

- **Data Extraction:** It collects the URL, Description (About), Programming Languages, Last Updated Time, and Number of Commits for each project.
- **Smart Logic:** * If a repository has no description, the script automatically uses the repository's name as the "About" text.
    - If a repository is empty (no commits), it smartly sets the Commits and Languages to "None".
- **Auto-Export to XML:** It neatly saves all the collected data into an XML file named repositories.xml.
- **Auto-Update (Overwrite):** Every time you run the script, it deletes the old file and writes the newest data, so your XML file is always up-to-date.

Chapter 2 Implementation

2.1 Class 1

This class inherits from scrapy.Spider. It acts as the main web crawler that visits the target GitHub profile, navigates through individual repository links, extracts the required project details, and applies the logical rules for empty repositories.

2.1.1 Fields

- name *(String)*: The unique identifier name for this spider (set to 'midterm_github').
- start_urls *(List)*: Contains the starting URL(s) where the spider begins crawling (the user's GitHub repositories page).
- custom_settings *(Dictionary)*: Contains Scrapy configurations, such as bypassing robots.txt, exporting the data format as XML, and allowing file overwrite.

2.1.2 Methods

- parse(self, response): The default method called by Scrapy to process the downloaded start_urls. It finds all repositories on the main page, extracts the basic data (URL, Repo Name, About, Languages, Last Updated), and sends a follow-up request to each repository's specific URL.
- parse_commits(self, response, my_data): A custom callback method used to process the individual repository pages. It extracts the commit numbers using regular expressions, checks if the repository is empty, applies the replacement logic for the "About" and "Languages" fields, and outputs the final data.

2.1.3 Functions

- re.search(): Used to find and extract the pure numbers (commit counts) from a raw string using Regular Expressions.
- bool(): Used to convert the commit text into a boolean value (True/False) to check whether the repository is empty or not.

2.2 Class 2

2.1 Method/Function 1

2.2 Method/Function 2

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Chapter 3 Results

3.1 Result 1

. <?xml version="1.0" encoding="utf-8"?>
<items>
<item><URL>https://github.com/Linx25059/ytDownloader</URL><About>Desktop App for downloading Videos and Audios from hundreds of sites</About><Languages>JavaScript</Languages><LastUpdated>2025-11-27T15:35:32Z</LastUpdated><NumberOfCommits>1022</NumberOfCommits></item>
<item><URL>https://github.com/Linx25059/mini-taiwan-pulse</URL><About>無說明</About><Languages>TypeScript</Languages><LastUpdated>2026-04-09T15:07:41Z</LastUpdated><NumberOfCommits>193</NumberOfCommits></item>
<item><URL>https://github.com/Linx25059/discord-social-preview-bot</URL><About>預覽社群的貼文！</About><Languages>JavaScript</Languages><LastUpdated>2026-04-13T08:58:59Z</LastUpdated><NumberOfCommits>12</NumberOfCommits></item>
<item><URL>https://github.com/Linx25059/andrej-karpathy-skills</URL><About>A single CLAUDE.md file to improve Claude Code behavior, derived from Andrej Karpathy's observations on LLM coding pitfalls.</About><Languages>未指定</Languages><LastUpdated>2026-04-15T17:47:20Z</LastUpdated><NumberOfCommits>25</NumberOfCommits></item>
<item><URL>https://github.com/Linx25059/group-5</URL><About>Fetch specified datas from GitHub.</About><Languages>Python</Languages><LastUpdated>2026-04-16T15:24:19Z</LastUpdated><NumberOfCommits>8</NumberOfCommits></item>
</items>

3.2 Result 2

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Chapter 4 Conclusions

In this project, we successfully developed an automated web scraping tool using Python and the Scrapy framework. The spider efficiently navigates a GitHub user's profile, accesses individual repositories, and extracts essential data such as URLs, descriptions, programming languages, update times, and commit counts.

Through this assignment, several key technical skills were demonstrated:

1. **Web Data Extraction:** Using CSS and XPath selectors to accurately target and extract specific HTML elements from dynamic web pages.
2. **Data Cleaning:** Applying Regular Expressions (Regex) to clean raw text and extract pure numbers (e.g., commit counts).
3. **Conditional Logic:** Implementing specific if/else rules to handle special cases, such as filling in missing descriptions with repository names and identifying empty projects.
4. **Automated Export:** Configuring Scrapy to automatically generate a well-structured XML file (repositories.xml) that safely overwrites older data upon every run.

Overall, this script provides a reliable, fast, and automated way to collect and organize GitHub repository data, successfully fulfilling all the requirements of the midterm assignment.